Paper Review • Embodied AI Datasets

Ropedia Xperience 2

A Gold Standard for Scaling Physical AI Environments and Behaviors

Real 3D scans × egocentric behavior: co-capture, registration, robot-data reuse

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One-Sentence Summary

Scan once, record many: reuse behavior in metric 3D scenes

> “One person LiDAR-scans a scene, then records tasks with HOMIE; video, poses, hands & body share one metric frame → **3D benchmark reference**, convertible to **sim & robot trajectories**.

1,589	4,230	74	131 h	596
Released scene scans	HOMIE recordings*	task definitions*	Total recording time*	capture sites*

* Collection-inventory counts; not guaranteed to be processed or registered. The paper has no scaling experiments: it answers “what data to scale and how to produce it,” not a scaling law.

LLMs control robots, yet remain brittle on long-horizon real tasks

22.48%

GPT-6 Astra • RoboDojo avg. success rate

0 / 11

Plug insertion: failed on all three platforms

9 / 23

Successful navigation + search trials

Two complementary resources are needed

Behaviors

Human demos of "how to do the task," plus derived robot actions and interactions

Environments

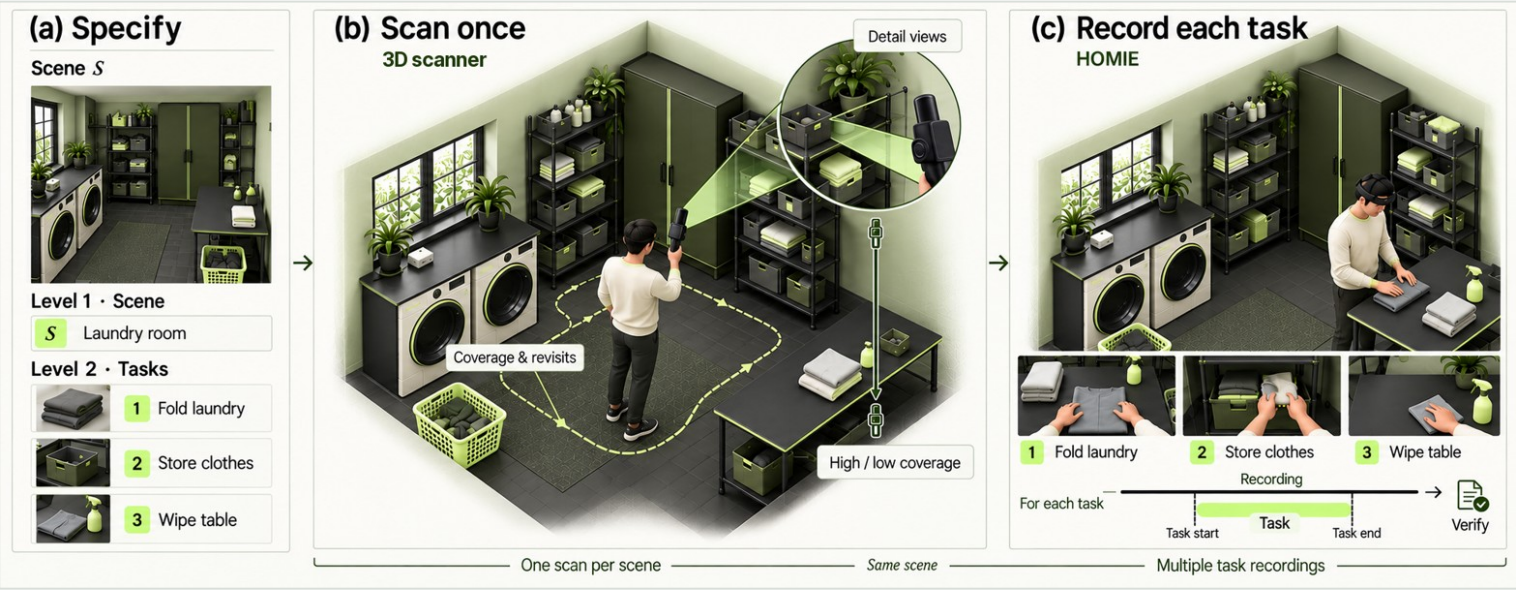
Real spatial layouts and object relations, enabling closed-loop policy rollout, evaluation and improvement

Core Question

How does physical AI capability scale with "environments" and "behaviors"?

Prerequisite: decide what data to scale and how to keep producing and reusing it → this paper's answer: "joint environment + behavior capture"

2 devices, 2 stages: one scan spatially grounds many recordings



3D Scanner • Handheld

LiDAR + 2 fisheye + 2 wide-angle (rolling shutter)

→ Metric mesh, 3DGS, COLMAP poses

HOMIE • Head-mounted

4 global-shutter fisheye (2 front, 2 rear) + IMU + 4-ch audio

→ Hands-free, follows egocentric view

Capture Tips

- Static: no screens/people/pets, fixed light, glass covered
- Scan: 2-way passes, close-ups, high/low views
- Record: whole task span; verify files after

Auto pipeline: observe → estimate → generate → metric scene frame

Observed

3D Scanner
Fisheye images

HOMIE
4 fisheye videos, 30 fps
IMU 200 Hz • 4-ch audio
HW sync < 50 µs

Estimated

Scan reconstruction
COLMAP poses • mesh • 3DGS (metric)

Stereo SLAM
6-DoF device pose + sparse map, 30 Hz

Stereo metric depth
Per-pixel confidence

Hands: MANO
21 joints/hand, stereo reprojection consistency

Body: SMPL-H
52 joints, EgoAllo (head-traj. cond. + hand guidance)

Generated

Task-level
Overall task & activity description

Subtask / action
With time intervals

Per-frame (2 fps)
Object state, hand state, hand–object interaction

Registration

Transforms

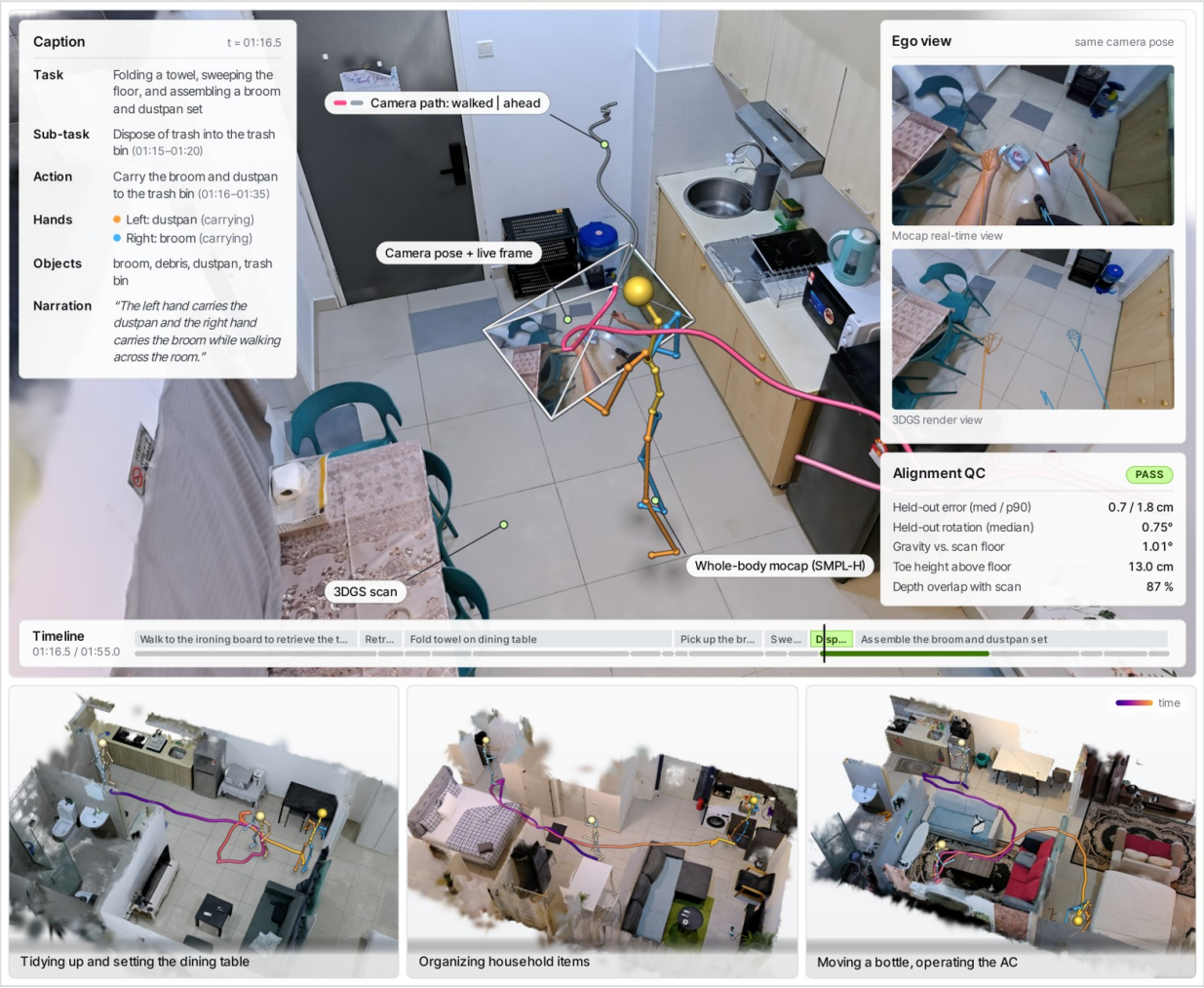
Recording SLAM frame

↓

Metric scene frame of scan

Trajectory, hands and body unified in scene; language labels kept with timestamps

Localize images + fit trajectory; independent geometric QC



1 **Localize** SIFT + RANSAC matching to scan images, keypoints lifted to mesh, PnP for camera pose

2 **Fit** Robust Umeyama, ~200 frames; even fit, odd held out; rescale only if span ⩾ 2.5 m

3 **Check** IMU gravity vs. floor normal; stereo depth vs. mesh ICP offset

PASS Held-out median ⩽ 5 cm, p90 ⩽ 12 cm; gravity ⩽ 2°; ICP ⩽ 10 cm / 2°

WEAK Fails any PASS criterion

FAIL No consistent init. / < 8 localized frames / p90 > 30 cm / ICP > 30 cm (e.g., same-layout neighbor)

Ex. PASS: 0.7/1.8 cm (med./p90), grav. 1.01°

Tasks span work sites, but recordings are heavily home-based

c Records by action and scene

RECORDS · LOG 1 — 1,000

		HOME					WORK / RETAIL		
		Bedroom	Bathroom	Kitchen	Living room	Laundry	Restaurant	Store	Office
PICK & PLACE	Place	887	54	109	140	1		20	
	Take out	786	13	111	125	2		12	
CLEANING	Wipe / clean	140	290	99	421	1	19	15	13
	Wash						22		
REARRANGEMENT	Arrange / tidy	29	69		210	1	26	13	10
FOLDING	Fold	460			127				
INSPECTION	Inspect							5	

74
task definitions

31
in work/retail settings

By action (no. of tasks)

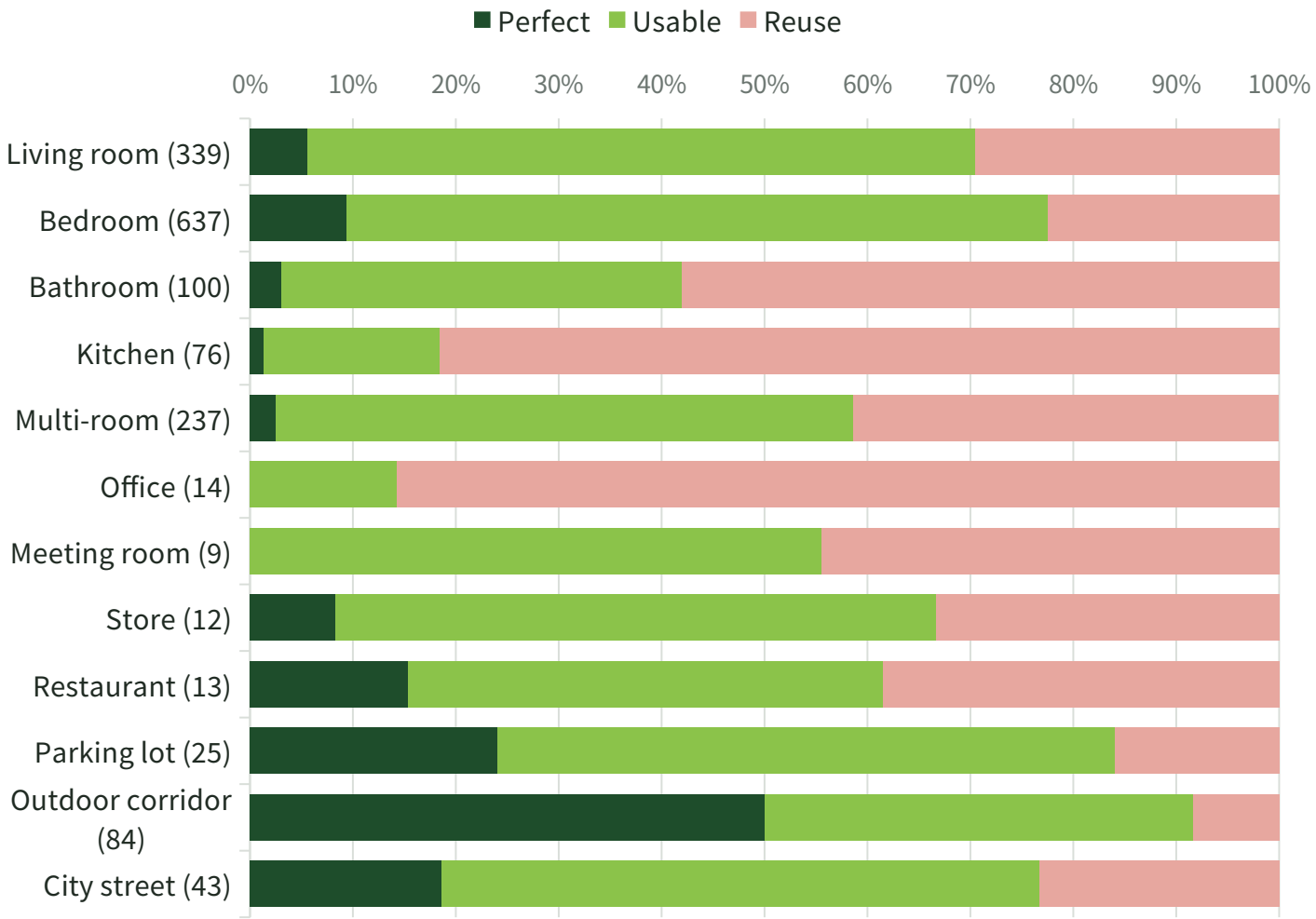
Place / remove 25 • Wipe/clean 23
Tidy 18 • Wash 5 • Fold 2 • Inspect 1

Most-recorded cells

Bedroom • Place **887**
Bedroom • Remove **786**
Bedroom • Fold **460**
Living rm. • Wipe **421**

Work/retail: non-empty cells have only 5–26 recordings

98.7% of scans kept, but only 9.3% Perfect; 82% of kitchens are Reuse



1,589
Kept (98.7%)

21
Delete （1.3%）

Perfect 148 • 9.3%

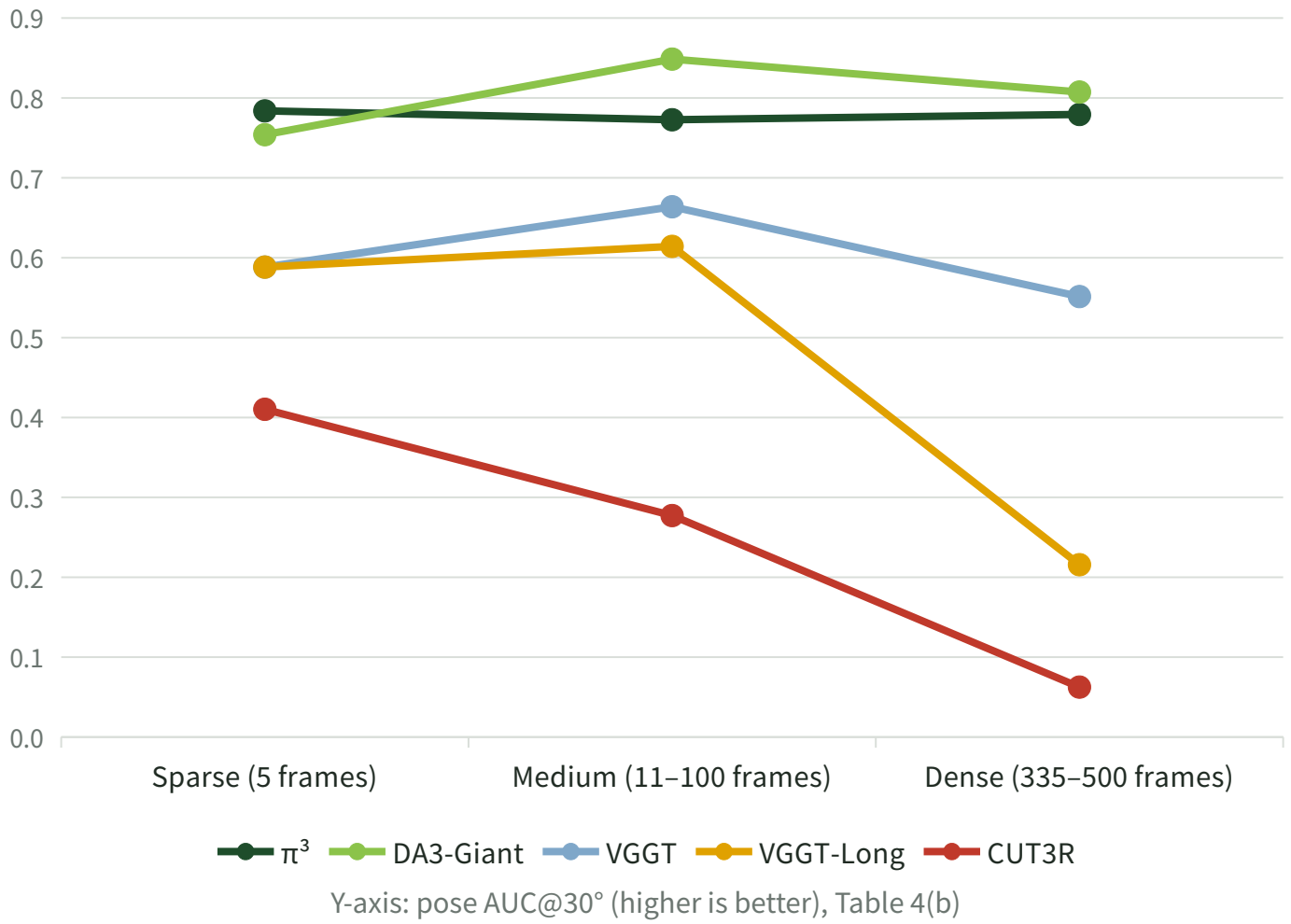
Usable 934 • 58.8%

Reuse 507 • 31.9%

Caveats

- Delete relabeled as Reuse in 8 scene types
- 91 scans without QC results kept as Usable
- 33 conflicting grades take the best
- Indoor home: 87.4%; bedroom + living room: 61.4%

Geometry: no all-rounder; more views not always better



Y-axis: pose AUC@30° (higher is better), Table 4(b)

π^3
Lowest ATE and highest F@5cm at all three densities (Sparse: ATE 0.43 m, F 0.25)

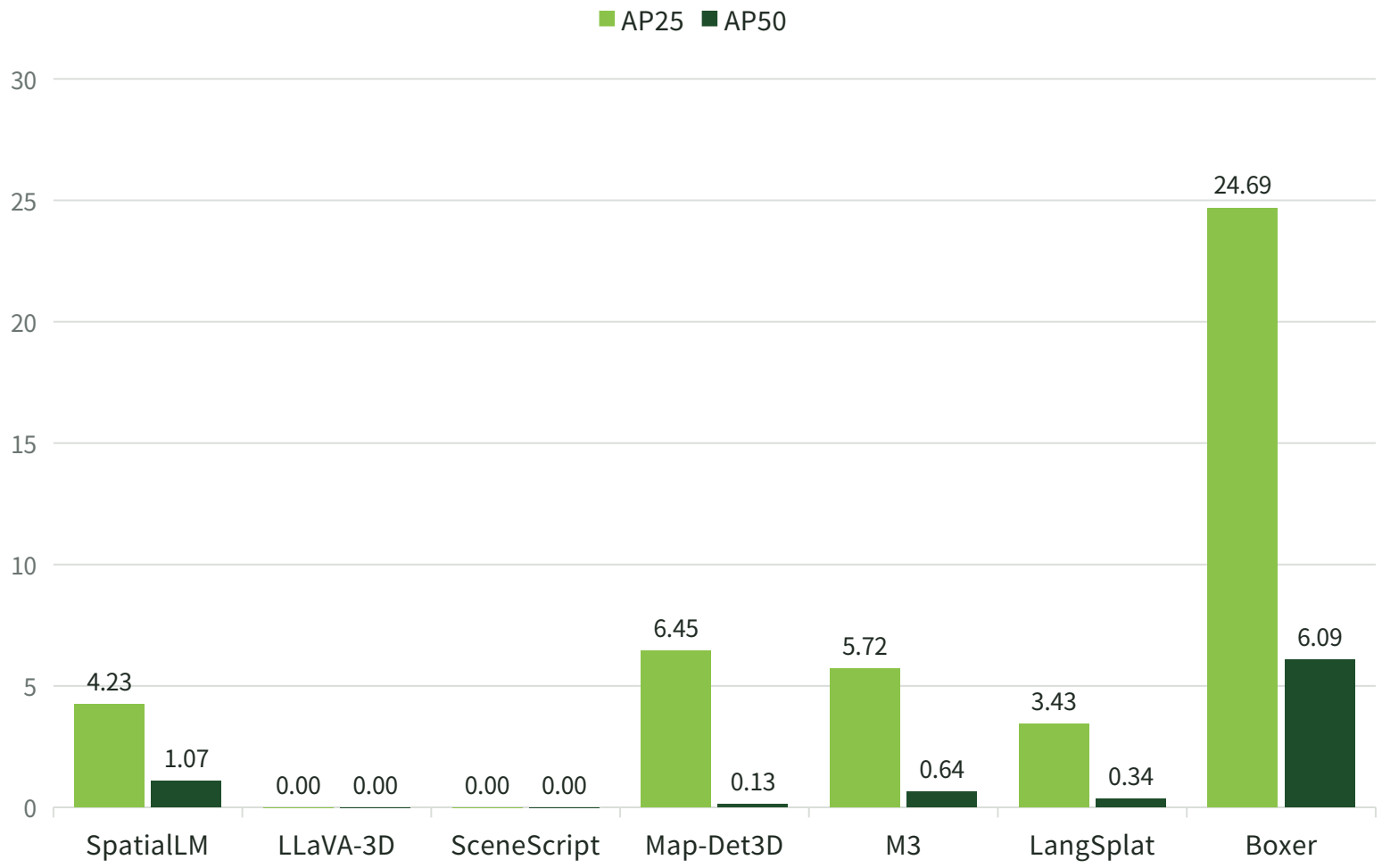
DA3-Giant
Highest Medium / Dense pose AUC: 0.849 / 0.807; best CD too

CUT3R
Pose AUC 0.41 → 0.06: worse with more views

DUSt3R / MASt3R
Out of GPU memory (OOM) on Dense input

Depth is median-scaled per scene, so quality is scale-normalized; reference geometry is scan pseudo-GT.

3D detection: tight 3D boxes in real clutter far from solved



24.69

Boxer AP25 (%) • best on all three

< 7

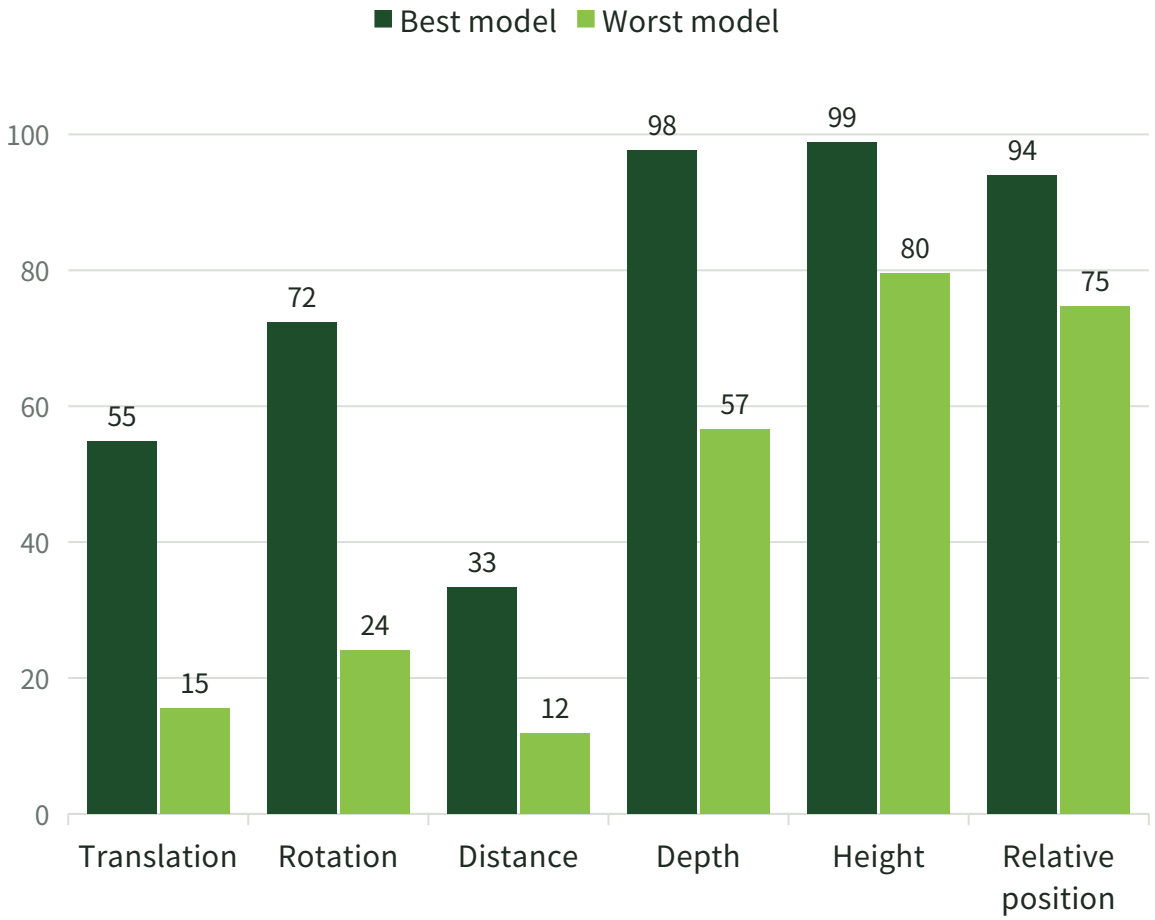
AP50 (%) of all methods

End-to-end: SpatialLM / LLaVA-3D / SceneScript / Map-Det3D

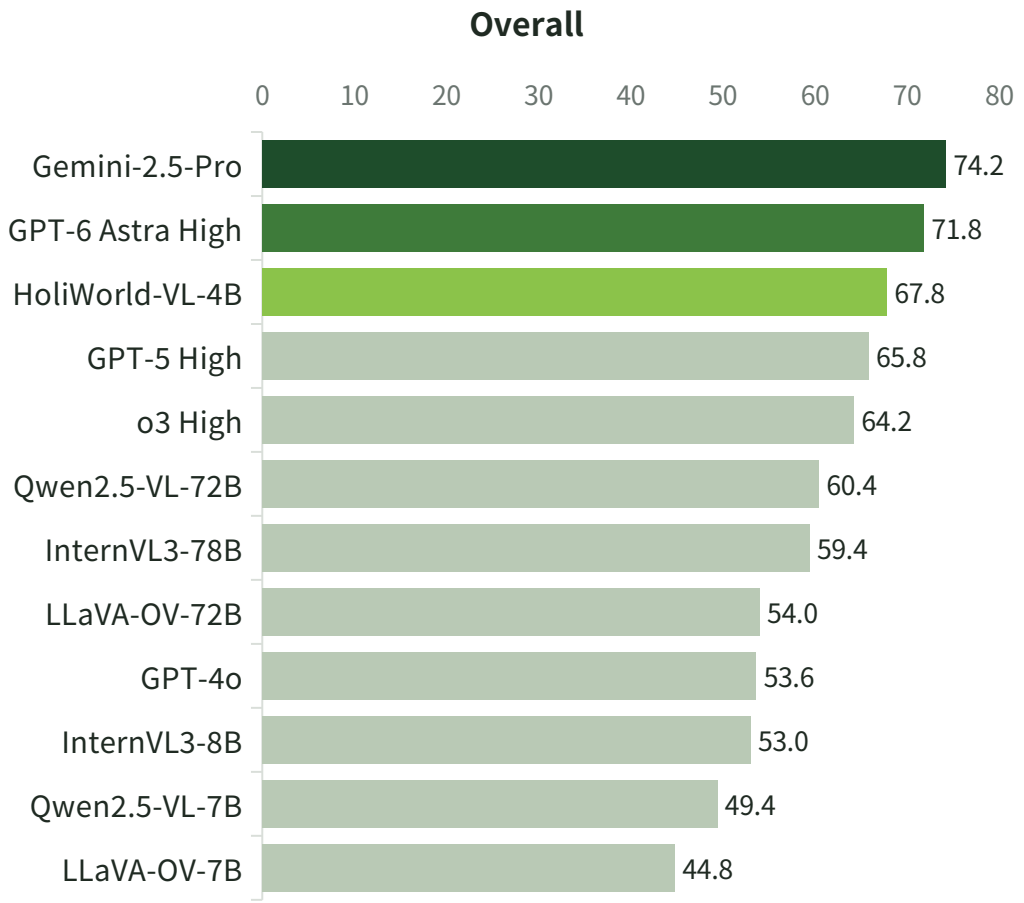
3DGS language fields: M3 / LangSplat

2D→3D lifting: Boxer

Spatial QA: height/rel. pos. easy; distance/translation widely hard

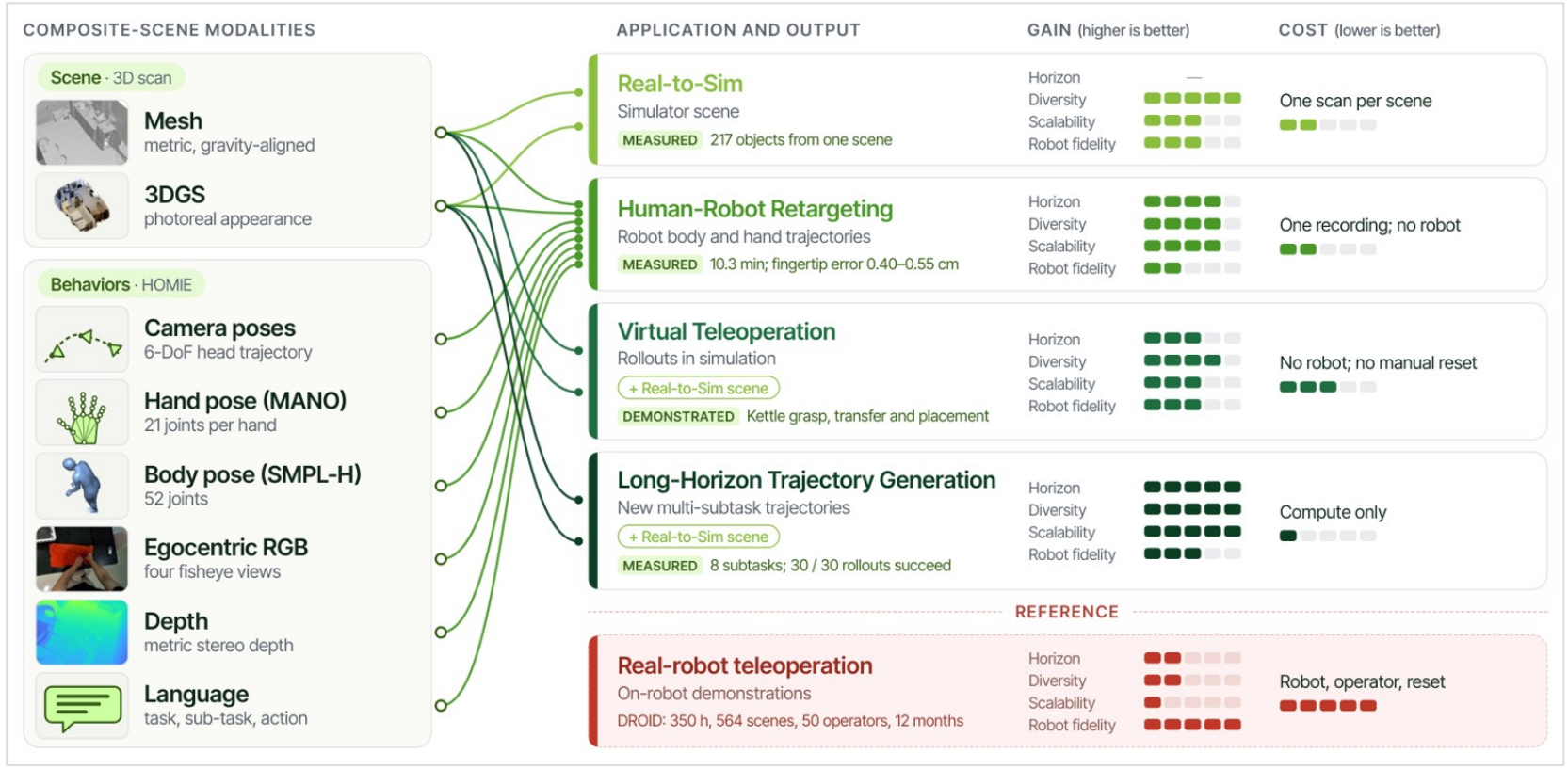


Highest and lowest accuracy (%) per question type



Overall accuracy (%); HoliWorld-VL-4B beats all open-source models (⩽ 60.4)

1 capture → 4 robot-data routes, scaling with operators, not robots



Scale Environments

Real-to-Sim: scan → interactive digital twin

Scale Behaviors

Human-to-robot retargeting

Virtual teleop (no real robot, auto-reset)

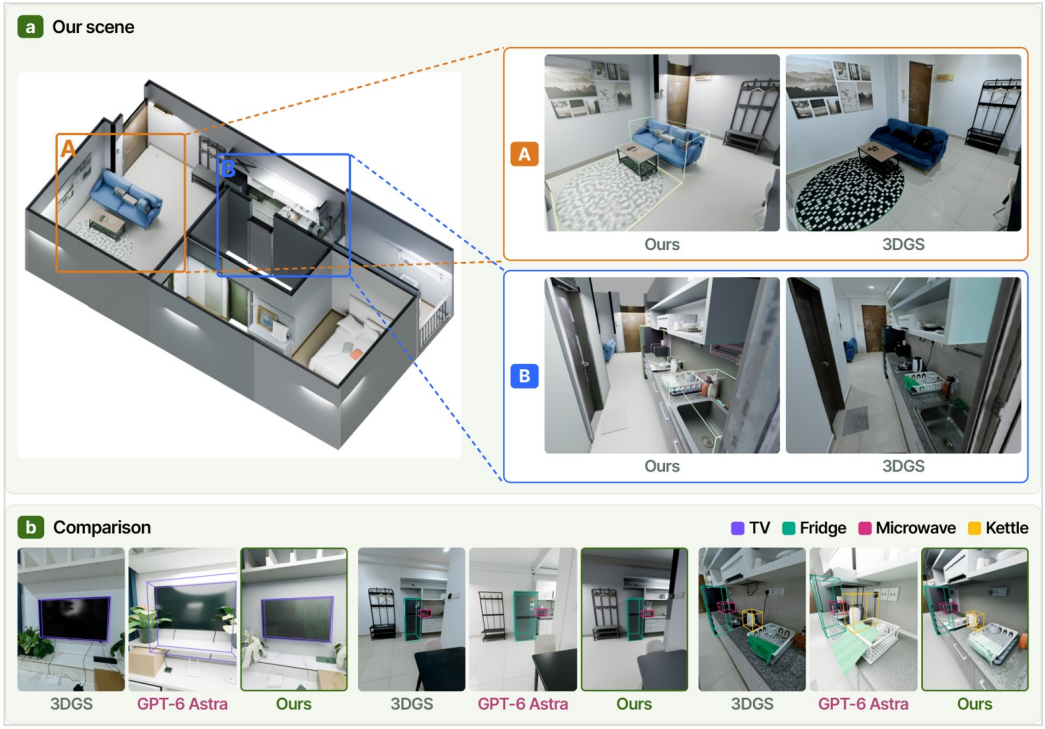
Long-horizon generation (compute only)

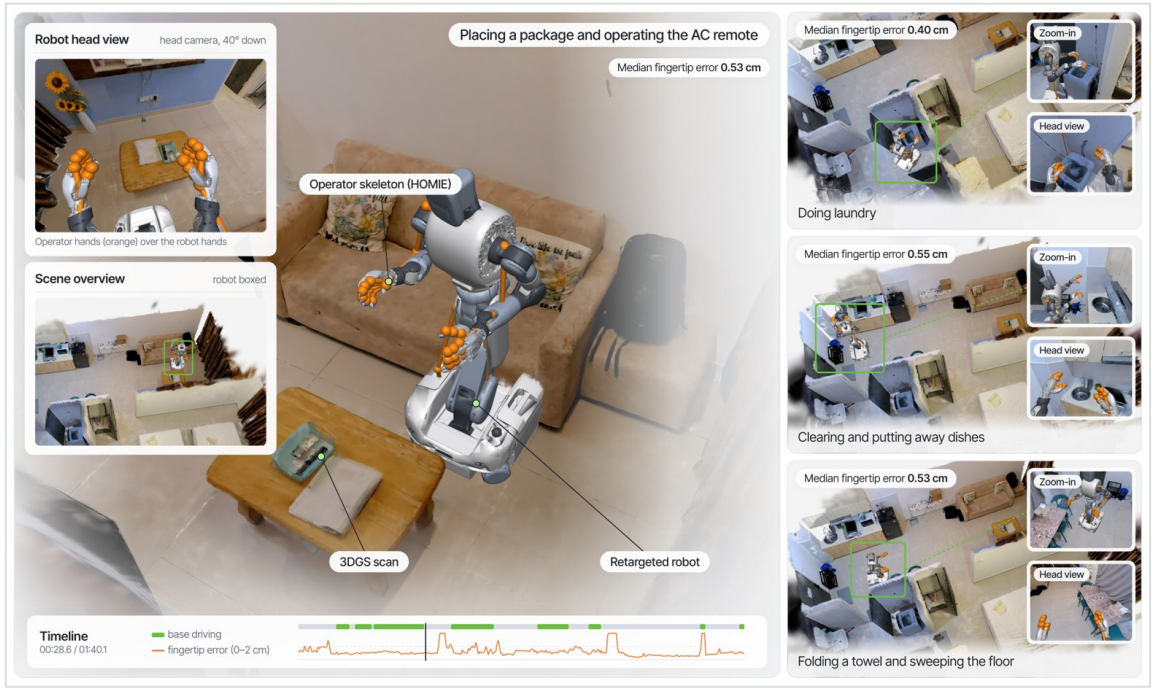
Reference: real-robot teleop

DROID 350 h • 564 scenes • 50 people • 12 months

Note: bars are qualitative 5-level ratings, not measurements

Real-to-Sim keeps metric geometry; retargeting tip error ≈ 0.5 cm





Real-to-Sim **217** distinct obj./scene

Per-view vs. 3DGS from same cameras: layout, size & appearance match; image-only build (GPT-6 Astra) drifts in color/size/position

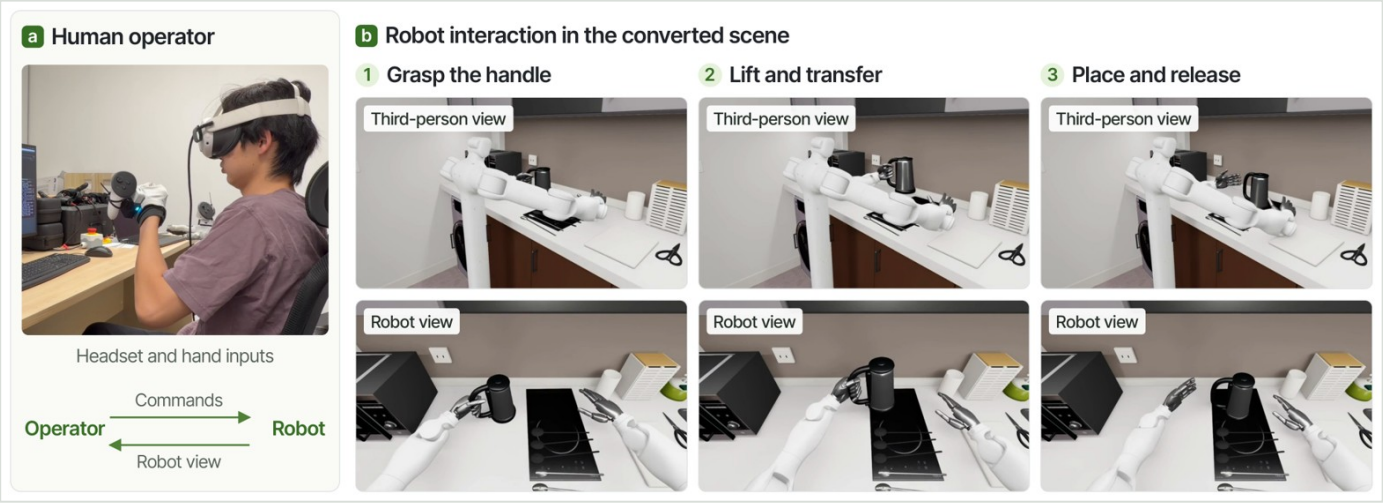
Boxes show 3DGS agreement, not independent GT accuracy

Retargeting **0.40–0.55 cm** median fingertip error

1 scene, 4 recordings • 10.3 min • base 96 m • p95 < 2 cm

All frames: joint vel./torque ⩽ 90% limit, ZMP in support poly., no collisions

Virtual teleop + long-horizon generation: 30/30 complete 8 subtasks

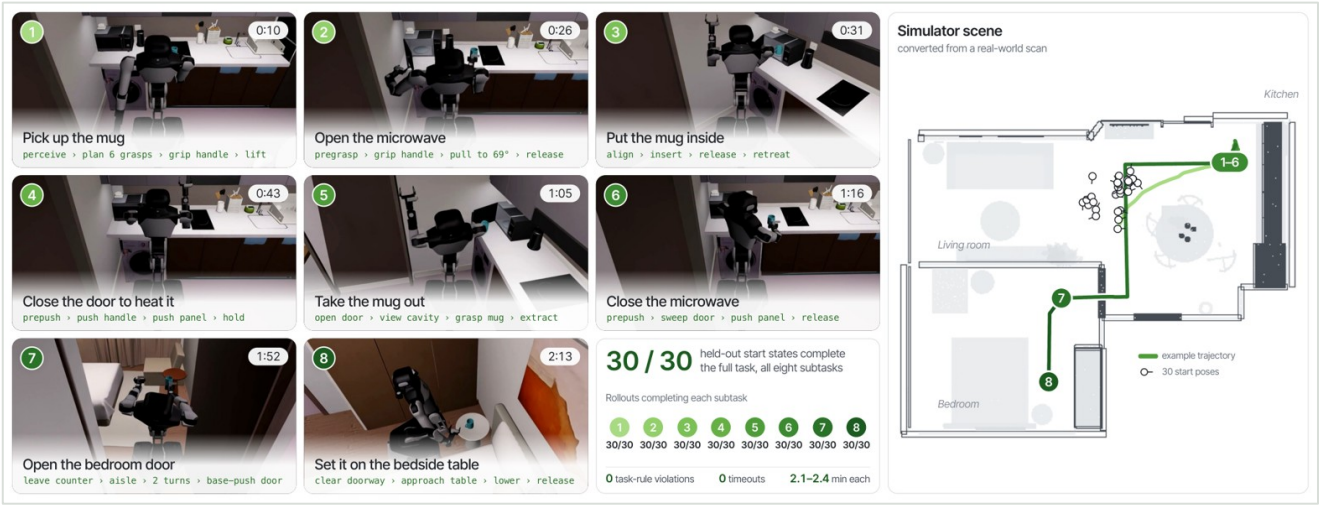


③ Virtual Teleoperation

Headset + hand input devices control a simulated humanoid in the converted kitchen:

Grasp kettle handle → lift & move → place on stove, release

No real robot; scene auto-resets in sim (single demo)



④ Long-Horizon Gen.

30/30 held-out starts do all 8 subtasks

No resets; 0 violations/timeouts; ≈ 2.2 min/episode; 10–16 min on 1 worker

Hand-coded policy; objects sensed only via perception; just 1 task

Solid, honest work; “gold standard” / “scaling law” untested

✓ Strengths

Practical, reproducible workflow

“Scan once, record many”: one person, no robot, no mocap needed for on-site capture

Transparent disclosure

Registration thresholds, held-out design and failure modes published; separates released/inventory and observed/estimated/generated

Informative benchmarks

Multi-view geometry: “more views not always better”; 3D detection AP50 < 7; VLM distance ⩽ 34%

! Reservations

No scaling experiments No policy trained on the data; no performance vs. environment/behavior scale

Robot demos are single instances 1 scene conversion, 1 teleop session, 1 long-horizon task (scripted policy)

Usable behavior volume unclear 4,230 is an inventory count; PASS rate not reported; 131 h ≪ predecessor’s 10,000 h

QC & benchmark scope Retention includes policy overrides; benchmarks use only 20 scenes, scan pseudo-GT

Distribution bias 87.4% indoor home; few work/retail recordings

Summary • Takeaways

A reproducible paradigm for “joint capture + spatial registration + reuse” of environments and behaviors

01

Data Paradigm

One person, two devices, no robot: 1,589 metric scans + egocentric behavior, unified in scene frames

02

Benchmark Findings

Geometry models not always better with more views; 3D detection AP50 < 7; VLM distance/translation reasoning broadly weak

03

Reuse Routes

Real-to-Sim, retargeting, virtual teleop and long-horizon generation all work end to end, but are single-instance demos so far

04

To Be Validated

The “gold standard” and physical-AI scaling laws need follow-up policy training and scaling experiments

arXiv 2611.17060 • ropedia.com/xperience/2-research • github.com/Ropedia/Ropedia-Xperience-2